

Curve Fitting & Interpolation Project

COE 311K – Engineering Computation

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Introduction

This project analyzes U.S. quarterly GDP growth rates between 2010 to 2023 using various numerical methods, including least squares regression, polynomial fitting, and cubic spline interpolation. The data is sourced from the U.S. Bureau of Economic Analysis (BEA). The goal is to compare interpolation and approximation techniques, evaluate their effectiveness in modeling real-world data, and recommend the most appropriate approach for economic data analysis.

Part A – Cubic Spline Interpolation

Method

To construct a natural cubic spline through the given data points, we create piecewise cubic polynomials between each pair of data points. For n data points, there are $n - 1$ cubic polynomials and $(4(n-1))$ coefficients. Each spline passes through all data points (interpolation condition), has first derivatives match at interior knots ($C1$ continuity), and has second derivatives match at interior knots ($C2$ continuity). Additionally, the second derivatives at the endpoints are set to zero, which defines the natural boundary conditions. This results in a

tridiagonal system of size $(n - 2) \times (n - 2)$, which is solved to determine the second derivatives at the interior points.

Evaluation and Plotting

The cubic spline was fit using only the 20 provided GDP data points as knots. The spline was evaluated at all integer quarter indices from 1 to 55 to produce interpolated values across the entire time range. The spline passes exactly through each provided data point and produces a smooth curve except in regions where the data changes abruptly, such as the quarters affected by the COVID-19 pandemic. The results are demonstrated in the plots below. For visualization purposes, there is one graph depicting the spline and interpolation behavior over the original 20 quarters data, and another graph demonstrating the spline evaluated over all 55 quarters.

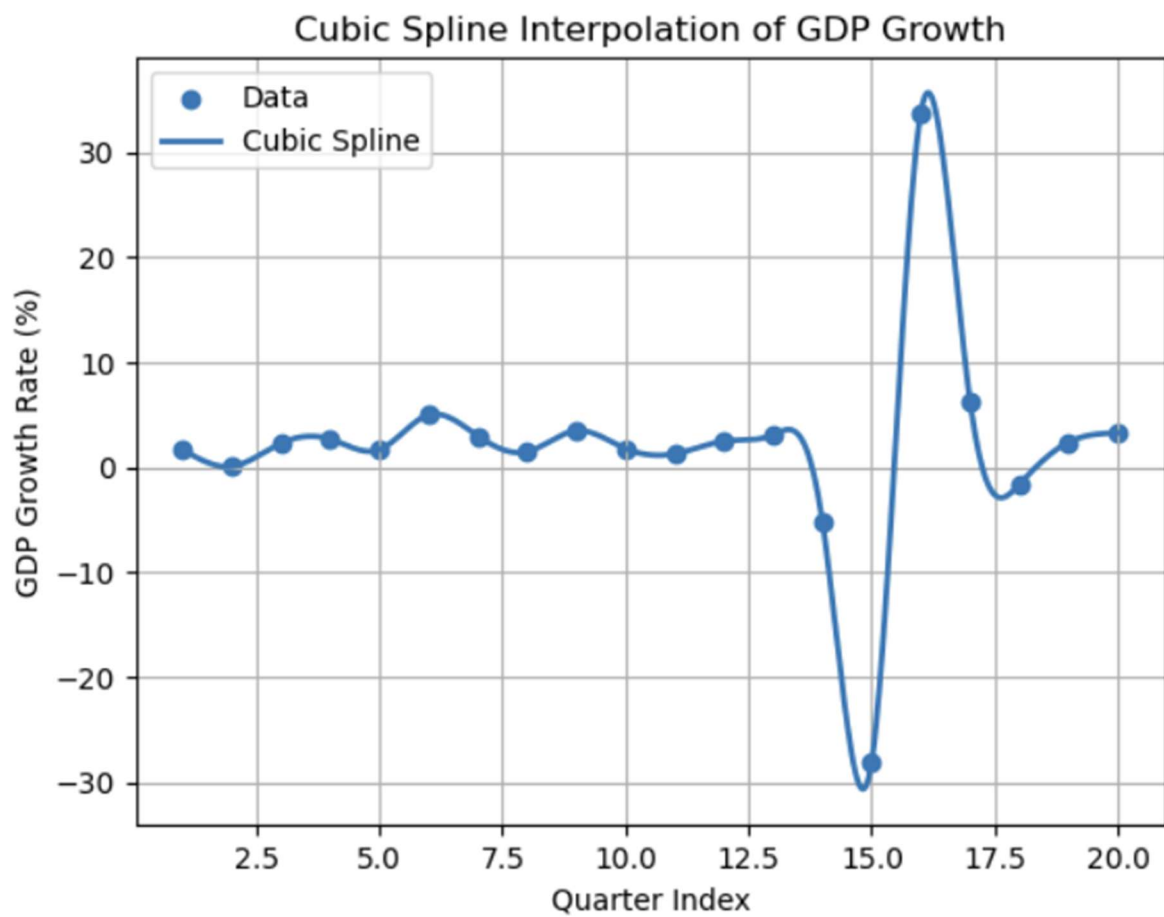


Figure 1: Natural Cubic Spline Interpolating 20 Quarterly GDP Growth Data Points (2010–2023)

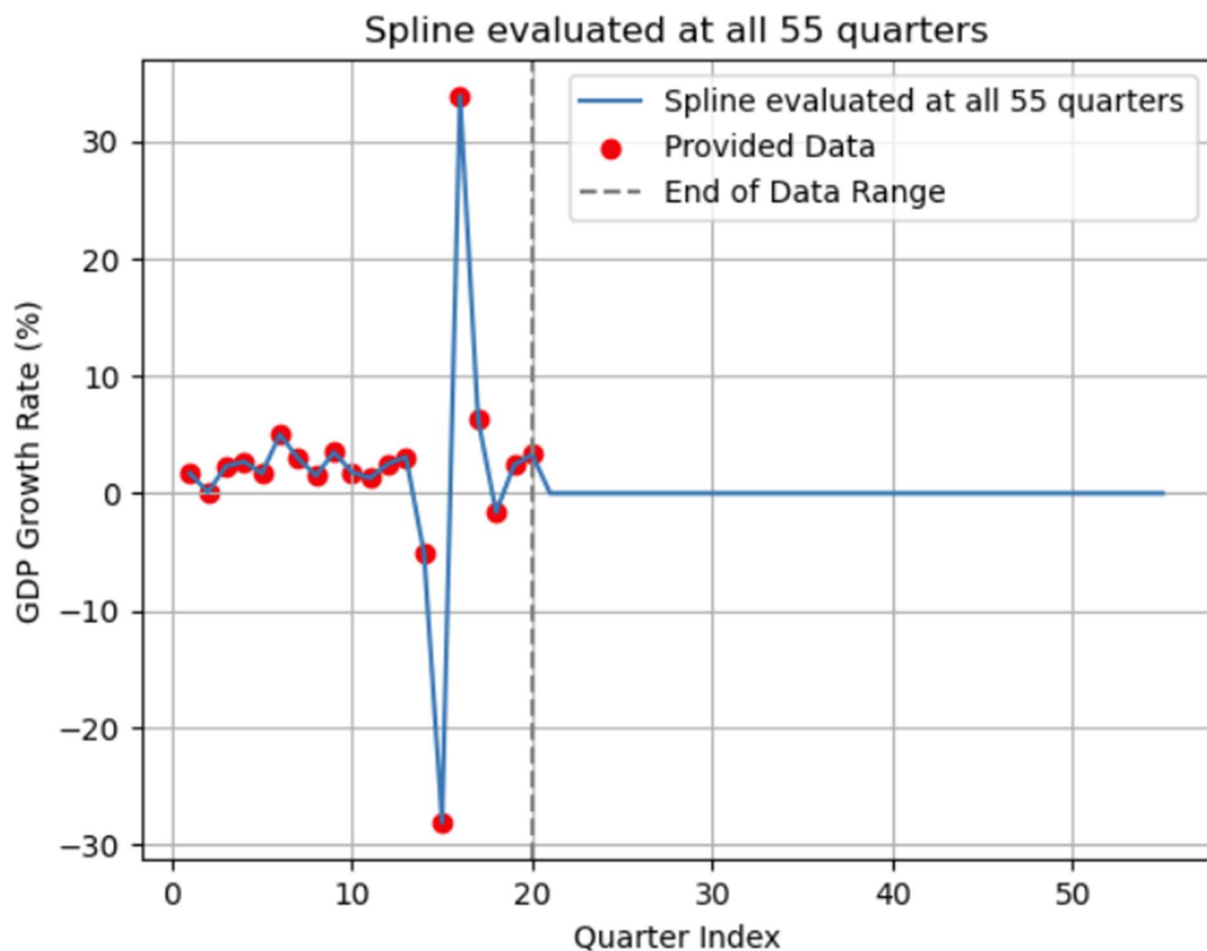


Figure 2: Cubic Spline Interpolation Evaluated Across All 55 Quarters

Results Discussion and Runge Phenomenon

The cubic spline is a reliable and precise way to fill in the gaps in the GDP growth data. It avoids big changes and keeps smooth transitions between data points by using low-degree polynomials locally instead of one high-degree polynomial globally.

The Runge phenomenon states that oscillations can occur when you use high-degree polynomial interpolation, specifically when you are close to the ends of a dataset or when there are extreme values. If a high-degree polynomial were used on this dataset, the significant changes during the COVID-19 period could cause big oscillations.

Cubic splines fix this issue by splitting the domain into smaller parts and fitting low-degree polynomials into each part. However, splines do pass through extreme outliers, which can locally distort the curve because they interpolate the data exactly. In these situations, smoothing techniques might be used to lessen the effect of extreme values.

Part B – Polynomial & Least Squares Comparison

The degree-4 polynomial fit shows the overall trend of the GDP growth data while smoothing out short-term changes. The polynomial doesn't go through every data point like the cubic spline does, which makes it less sensitive to extreme values like those seen during the COVID-19 pandemic.

The cubic spline, on the other hand, interpolates all data points perfectly, so it reacts more quickly to sudden changes in the data. This makes it very good at interpolation, but it also makes it more likely to pick up on outliers.

The linear regression model gives a rough idea of the overall trend, but it doesn't show how the data changes. The residual plots show this clearly: the linear model has bigger and more organized errors than the polynomial fit.

The RMSE values also support this conclusion because the polynomial model makes smaller errors than the linear model, which means it fits the data better.

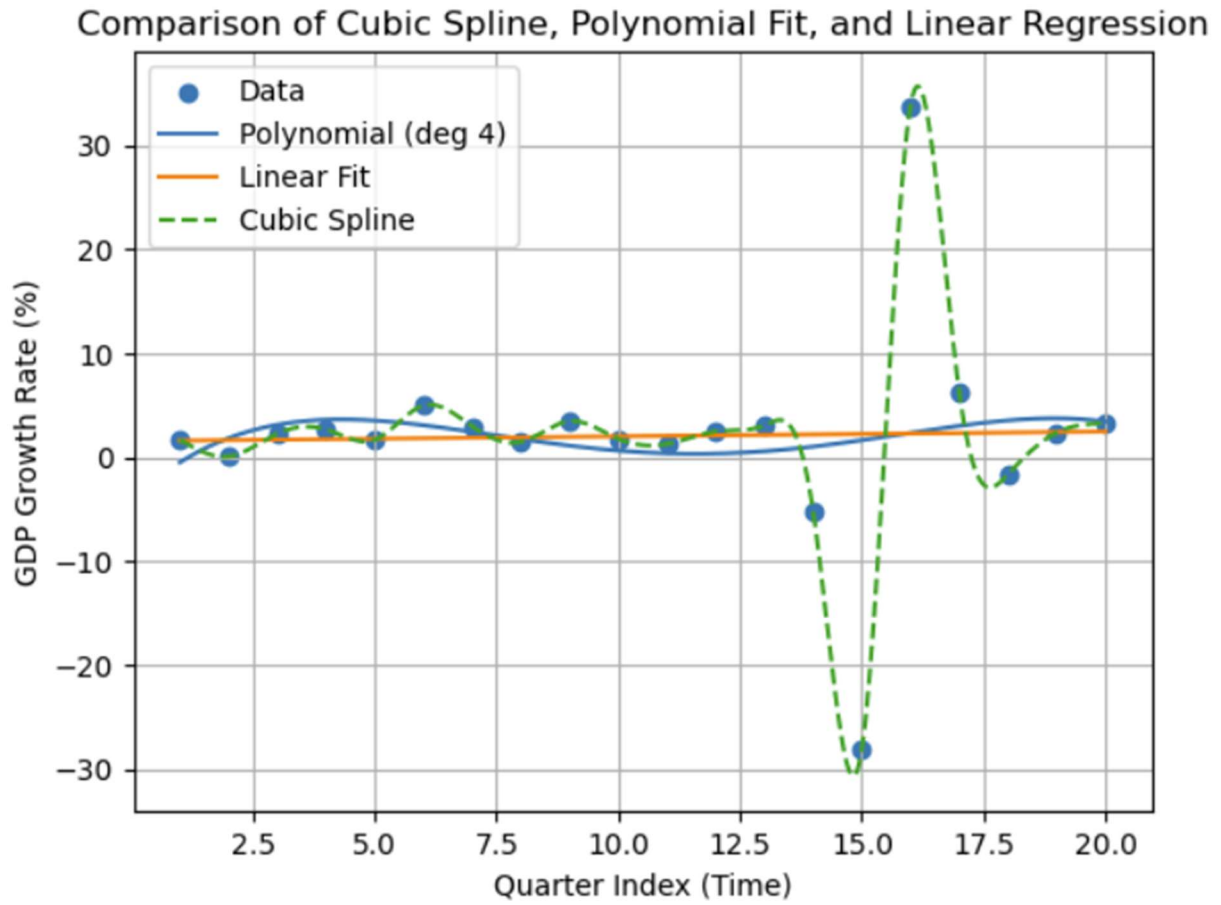


Figure 3: Comparison of Degree-4 Polynomial, Cubic Spline, and Linear Regression Fits to GDP Growth Data

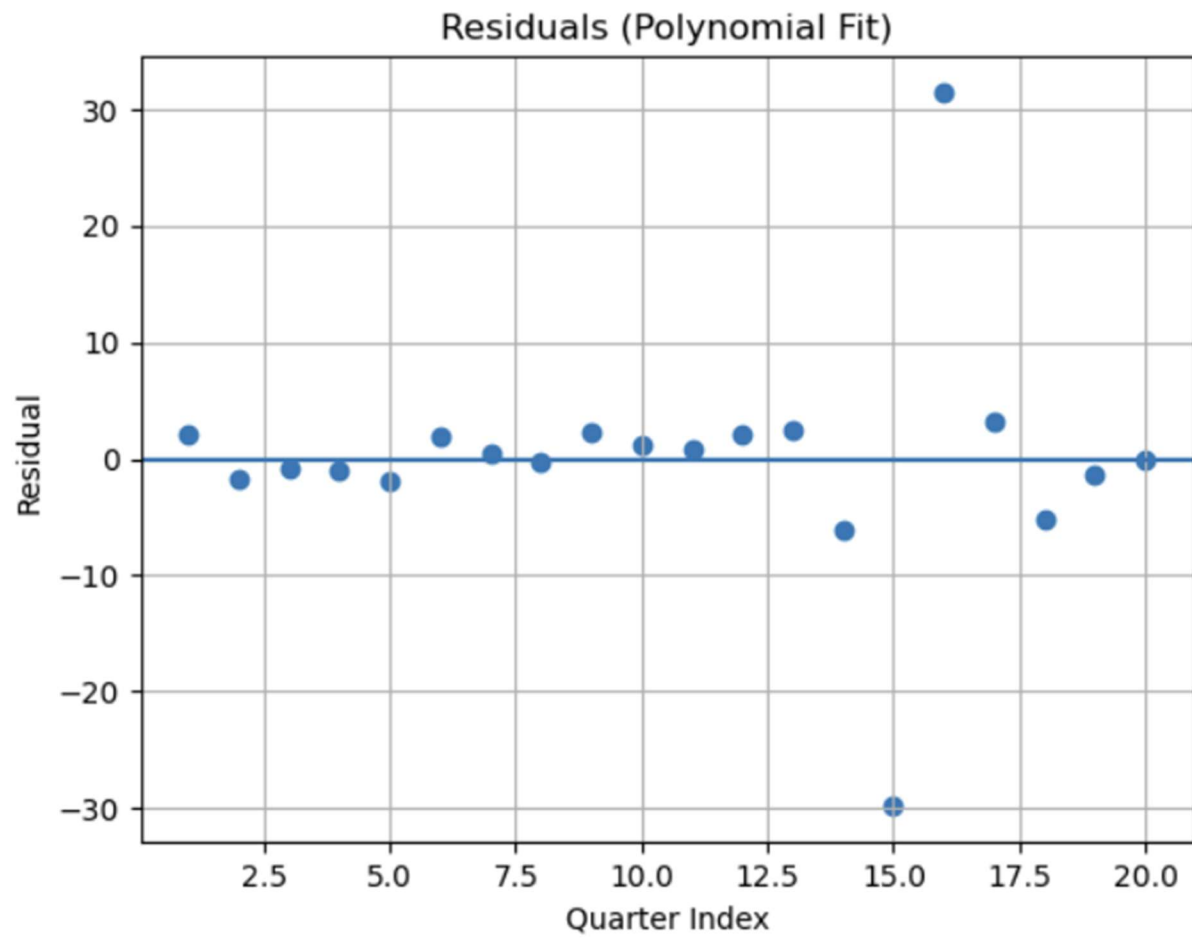


Figure 4: Residuals of Polynomial, Cubic Spline, and Linear Regression Models (Polynomial Fit)

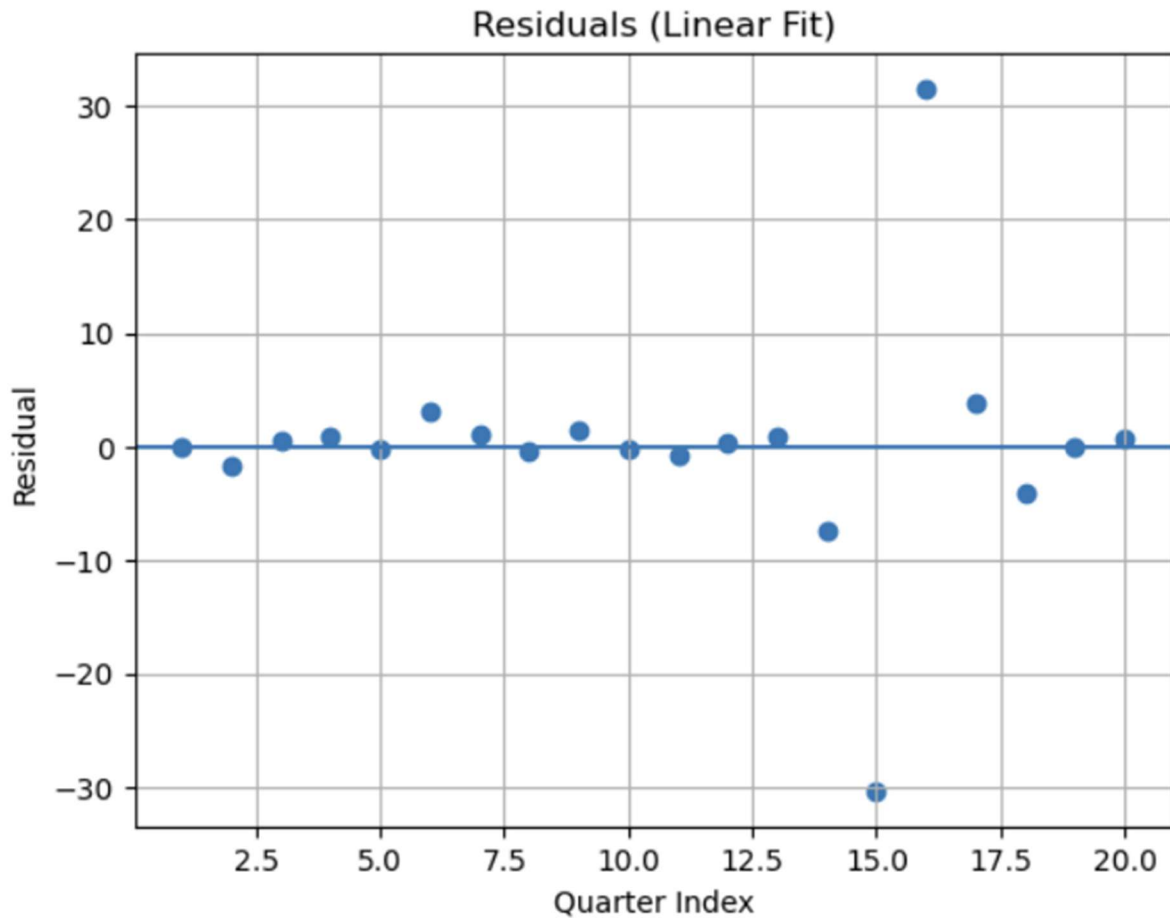


Figure 5: Residuals of Polynomial, Cubic Spline, and Linear Regression Models (Linear Fit)

RMSE (Polynomial): 9.965326786519533
 RMSE (Linear): 10.045790874078271

Figure 6: RMSE Values for Polynomial and Linear Fits

Part C – Method Justification

Cubic spline interpolation is the best way for a policymaker to figure out how much GDP will grow between two known data points. The spline makes a smooth curve with first and second

derivatives that are always changing. This makes sure that the transitions between values are realistic while still matching known data points perfectly.

Polynomial fitting is helpful for finding general trends, but it doesn't guarantee accuracy at specific points and can even cause oscillations, especially when there are outliers. This dataset is too simple for linear regression, and it doesn't show important changes in GDP growth.

Big O Analysis

Big O notation tells us how the time it takes for an algorithm to run changes as the input data gets bigger. The methods used in this project—linear regression, polynomial fitting with a fixed degree, and cubic spline interpolation—all have a time complexity of $O(n)$, where n is the number of data points. In this case, all methods scale linearly, so the choice of method isn't heavily influenced by how efficient it is to compute. Instead, the choice is based on how accurate, smooth, and able to handle real-world data features like noise and outliers the system is.

Conclusion of Justification

Cubic splines are the best way to fill in the gaps between known GDP data points because they are both very accurate and smooth without making things unstable. The polynomial and least squares methods are better for finding long-term trends than for guessing specific intermediate values.

Conclusion

This project used cubic spline interpolation, polynomial fitting, and least squares regression to look at U.S. GDP growth data. Each method showed its own strengths based on what it was

meant to do. Cubic splines were the best for interpolation because they go through all the data points exactly and make a smooth, continuous curve. Polynomial fitting was a good way to get a rough idea of the overall trend while making it less sensitive to noise. Linear regression, though straightforward, failed to account for the variability inherent in the dataset.

This project shows how important it is to choose the right numerical methods for the problem at hand. Interpolation techniques are better for figuring out values between known data points, while approximation methods are better for figuring out general trends in real-world data.